

Sharp identifiability boundary for level-2 Jukes–Cantor networks

Status

PROVED. All claims in this report are under the standard semi-directed reduction and the open JC parameter domain

$$0 < x_e < 1, \quad 0 < \lambda_r < 1.$$

The three tree-child classes are kept distinct:

- R_{TC} : the supplied rooted DAG is tree-child;
- W_{TC} : the standard semi-directed topology has at least one tree-child rooted partner;
- S_{TC} : every admissible rooted partner is tree-child.

The network class consists of binary standard semi-directed S_{TC} level-2 networks with at most one triangle in each blob.

Primary theorem

Theorem A — complete one-sided observational classification

PROVED. Let N, N' be leaf-labelled networks in the stated class. Then

$$N \preceq_{\text{JC}} N'$$

if and only if their labelled homeomorphism-reduced bridge trees agree and, under the induced correspondence of ports, every pair of corresponding blobs is either labelled-isomorphic or differs only by ordinary triangle redirection T .

Every surviving local relation is in fact a symmetric regular overlap. Thus ordinary triangle redirection is the complete observational ambiguity in S_{TC} .

Corollary B — generic identifiability modulo T

PROVED. For each fixed S_{TC} topology N , there is a proper algebraic exceptional set in its parameter space such that every parameter outside that set is identified by its exact JC distribution up to ordinary triangle redirection and labelled isomorphism.

Theorem C — sharpness

PROVED. The conclusion fails on replacing S_{TC} by W_{TC} . For every $n \geq 4$, the inherited Theta construction gives nonisomorphic, non- T -equivalent networks

$$N_n, N'_n \in W_{\text{TC}} \setminus S_{\text{TC}}$$

whose open JC model images share a full-dimensional regular region. Their model dimension is $2n$.

Consequently, the boundary is sharp:

S_{TC} is generically JC-identifiable modulo T , W_{TC} is not.

1. The final seven-port universe

The inherited conservative completion census contained 192 records with three additional rigid-support labels. The stale prose field saying that core 3 required at most two additions has been corrected. The explicit machine-readable distribution had already recorded

$$(\text{core 3, 3 additions, 7 outgoing ports}) = 192.$$

A deterministic replay of the weak-presentation indexing reconstructs these records as eight role patterns, each with all 24 outgoing-label permutations:

orbit id	selected path-sink?	ordinary port counts on (UV, SU, SV, UX, VX)
629	no	(0, 0, 1, 1, 2)
644	no	(0, 1, 1, 2, 0)
649	no	(0, 2, 0, 1, 1)
650	no	(0, 2, 0, 2, 0)
685	no	(2, 1, 0, 1, 0)
700	yes	(0, 0, 2, 0, 1)
705	yes	(0, 1, 0, 2, 0)
706	yes	(0, 1, 1, 0, 1)

The directed core has vertices

$$S : \text{source}, \quad U : \text{tree branch}, \quad V : \text{branch reticulation}, \quad X : \text{path-sink reticulation},$$

and segments

$$U \rightarrow V, \quad S \rightarrow U, \quad S \rightarrow V, \quad U \rightarrow X, \quad V \rightarrow X.$$

Its minimum strong repairs are $\{UV, VX\}$ and $\{UX, VX\}$. These vertex roles make the directed-core automorphism group trivial. Hence the S_4 action is free on each role pattern, the eight orbits are disjoint, and all 192 records occur exactly once. The certificate lists every canonical graph hash.

For the eight canonical representatives, every conservative completion by labels 5, 6, 7 was enumerated. There are 1,686 distinct completed labelled mixed graphs. Every one satisfies the standard S_{TC} tail criterion: each tail of a retained reticulation arrow has two incident undirected edges. The complete displayed-tree monomial schema of every completion is hashed in the certificate.

2. Universal three-port separator

For three selected ports a, b, c , put

$$F_{abc} = r_{ab}r_{ac}r_{bc} - u_{abc}^2,$$

where r_{ab}, r_{ac}, r_{bc} are the three pair coordinates and u_{abc} is the all-distinct zero-sum Fourier coordinate.

For a cycle model with path-sink label s :

$$F_{abc} = 0 \quad \Longleftrightarrow \quad s \notin \{a, b, c\}.$$

If $s \in \{a, b, c\}$, the induced three-port cycle has one of two forms. Exact contraction gives

$$F = \lambda(1 - \lambda)M(1 - y)^2 > 0$$

when the two ordinary ports lie on one side, and

$$F = \lambda(1 - \lambda)M(1 - yz)^2 > 0$$

when they lie on opposite sides, where M is a positive monomial. Thus the zero/nonzero pattern of the ten three-subsets is exactly a six-edge star centred at the cycle sink.

Six orbits separated before completion

PROVED. The exact pullbacks on the original four outgoing ports plus the incoming port give:

- 644, 650, and 685: all ten F 's vanish;
- 629: target-zero triples containing every possible cycle-sink choice;
- 700 and 706: for each possible cycle sink, a target-positive triple omitting that sink.

Hence all 144 labelled records in these six orbits are stochastically disjoint from the proposed cycle source without using the extra labels.

The two genuine completion cases

Only 649 and 705 have the cycle-star pattern on the original five ports. It forces the cycle sink to be old label 4.

For orbit 649, every completion has a newly selected target sink $e \in \{5, 6, 7\}$. On the complete target tensor, marginalize to $(1, 2, e)$. Exact contraction yields

$$F_{12e} = M_A(1 - \lambda_V)(1 - \lambda_X)(1 - a)^2 \\ \times [\lambda_V(1 - \lambda_X)b + \lambda_Xc] > 0,$$

with M_A, a, b, c positive products of open edge multipliers. The cycle source has sink 4, so $F_{12e} = 0$ there.

For orbit 705, every strong completion must put at least one new label $e \in \{5, 6, 7\}$ on segment $V \rightarrow X$. Marginalization to $(1, 2, e)$ gives

$$F_{12e} = M_B\lambda_V(1 - \lambda_V)(1 - a)^2 > 0,$$

while the cycle source again has $F_{12e} = 0$.

The verifier expands both identities from the full seven-port displayed-tree parameterizations. Extra subdivisions merely replace the displayed symbols by products in $(0, 1)$, so strict positivity is unchanged.

Therefore every one of the 192 residual relations has disjoint complete open stochastic images. No isolated-point or closure-only argument is used.

3. Exact local dimensions

For the lifted incoming-port parameterizations used in the seven-port certificate, the root lies outside the local factor, its two root edges are ordinary, and the three-edge neighborhoods of distinct reticulations are disjoint. Such a binary rooted parameterization on n observed boundary leaves with r reticulations has

$$E = 2n + 3r - 2$$

edges and $E + r$ raw JC parameters. Its Fourier parameterization has one root-split gauge and two independent gauges per reticulation.

Indeed, if a reticulation has incoming multipliers a, b , outgoing multiplier c , inheritance λ , and descendant character sum h , then:

- when $h = 0$, switching the selected parent changes upstream descendant sets only by a zero character sum, so the two contributions combine with weight $\lambda + (1 - \lambda) = 1$;
- when $h \neq 0$, the local coefficients are exactly λac and $(1 - \lambda)bc$.

Thus four local raw parameters enter through two combinations. Consequently

$$\dim V \leq E + r - (1 + 2r) = 2n + 2r - 3.$$

For seven outgoing ports plus the incoming port, $n = 8$. Hence

$$\dim V_{\text{cycle}} = 15, \quad \dim V_{\text{core3}} = 17.$$

The certificate supplies a nonzero exact Jacobian minor modulo the prime 1,000,003 for every one of the seven cycle side-count vectors and all 65 core-3 count vectors occurring among the 1,686 completions. An independent implementation re-evaluates every stored minor. These lower bounds meet the structural upper bounds exactly.

The dimension difference does not by itself rule out cycle-to-theta containment. The strict F separators do.

4. Arbitrary-subdivision promotion

PROVED. The bounded-support reconstruction theorem supplies, in every arbitrarily subdivided local factor:

1. a rigid support containing the path-sink children and one port on each segment of a minimum strong repair;
2. one-port probes locating every remaining port's directed segment;
3. two-port probes recovering the order of every pair on a segment.

For cycle factors the corresponding support consists of the sink child and one side anchor, with one- and two-port probes recovering the two ordered sides.

If an arbitrary cycle source were one-sided contained in a theta target, its bounded restrictions would enter the completed cycle/theta atlas. All cases through six outgoing ports were already closed; the only omitted completion was precisely the 192-record core-3 universe above. Since all 192 now have a strict open-domain separator, no cross-generator containment survives.

The canonical bounded decks use every ordered four-port restriction of the complete joint boundary tensor: $6P4 = 360$ features at five outgoing ports and $7P4 = 840$ at six. Consequently the computational choice of an incoming port is auxiliary. Rerooting or changing that auxiliary designation merely permutes the full joint-tensor coordinates, and the all-relative-labelling atlas transports with that permutation.

The previously completed theta/theta, cycle/cycle, and theta-source/cycle- target directions, together with ordered-word reconstruction, therefore give:

Nonroot local theorem

PROVED. For arbitrary finite port-labelled nonroot S_{TC} level-2 cycle or theta factors with at most one triangle,

$$B \preceq_{\text{JC}} B'$$

occurs exactly when B, B' are port-labelled isomorphic or related by ordinary triangle redirection T .

5. Root reduction

PROVED. In any tree-child rooted partner, start at the root and repeatedly choose a tree-or-leaf child. Acyclic finiteness produces a root-to-leaf path containing only ordinary tree vertices and the terminal leaf.

Suppress the old root, reverse the ordinary edges on this path, and insert a new root on the terminal pendant edge. Binary degrees are preserved, reticulation arrowheads are unchanged, and no directed cycle is created. The new rooting is admissible; because the topology lies in S_{TC} , it is tree-child. Standard reduction forgets the reversed ordinary directions and therefore returns the identical semi-directed topology.

For JC, uniform stationarity and reversibility make every displayed-tree distribution root-independent. There is one directed-edge case worth making explicit. If the old root's off-path child is a reticulation, then in a switching selecting that parent the two old root-arm multipliers occur only through their product. In a switching selecting the other parent, the remaining one-child root stem subtends all leaves, has total character zero, and contributes exponent zero. Thus suppression again gives exactly one effective retained-edge multiplier.

If relocation splits any effective multiplier x , choose

$$y = \frac{1+x}{2}, \quad z = \frac{2x}{1+x}.$$

Then $0 < y, z < 1$ and $yz = x$. Conversely, multiplying the two old root arms produces an open effective multiplier, so the two root placements have the same complete JC image germ.

The resulting boundary tensor is a joint tensor; which boundary was used as the computational incoming port is not observed. By the all-ordered-port covariance of the bounded atlas, every root-local factor is therefore covered by the nonroot local theorem even when two compared topologies use different admissible pendant root sites.

6. Pointwise cut preservation

The frozen pointwise certificate contains 177 canonical three-boundary endpoint tensor types. Every type obeys exactly one branch:

$$abc - t^2 > 0$$

for 151 types, or

$$abc - t^2 = 0, \quad a - bc > 0$$

for 26 types.

For a crossing contained in one active local factor, the complete four-port universe has 453 types. Of these, 421 possess a fixed-sign wrong-split minor; the remaining 32 have all wrong-split blocks rank one and contain the tested bridge in every switching.

For a bridge joining two active endpoint tensors, write their coordinates as (a, b, c, t) , (A, B, C, T) , and the bridge multiplier as $z \in (0, 1)$. Four exact wrong-split minors are

$$\begin{aligned} m_0 &= aA - bcBCz^2, \\ m_1 &= (aA - Ttz)(aA + Ttz), \\ m_2 &= aA^2 - bcT^2z^2, \\ m_3 &= a^2A - t^2BCz^2. \end{aligned}$$

Rank one would imply

$$abc = t^2, \quad ABC = T^2, \quad aA = bcBCz^2.$$

The endpoint dichotomy then gives $a > bc$ and $A > BC$, so $aA > bcBC$; the third equation and $z < 1$ give $aA < bcBC$, a contradiction.

Hence, throughout the open JC domain, a flattening has rank at most four exactly at a cut split. One-sided open containment therefore preserves cuts in both directions: a cut on one side and non-cut on the other would require the same common distribution to have flattening rank both at most and greater than four. The labelled bridge trees agree.

7. Global synthesis

Recover the common bridge tree and peel its Fourier rank-one blocks. This extracts every local tensor up to one positive reciprocal JC gauge per bridge. The audited local invariants are multihomogeneous in the port arms, so these gauges preserve their zero and sign tests.

Fix positive anchor entries in every rank-one cut block. They give an analytic gauge slice in which a regular source model germ is a Cartesian product of its local tensor germs and open bridge parameters. The same factorization describes the target set; it does not require a regular target preimage. If a source-relative product box lies in the target model, then projection of that box to each coordinate factor is a source-relative open one-sided containment of the corresponding local model. Root reduction and the nonroot local theorem therefore force every local pair to be isomorphic or T -related.

Conversely, an ordinary triangle in a valid standard S_{TC} factor has one reticulation and three always-present ordinary external arms. Cutting beyond those arms factors the network map through the same normalized four-dimensional three-boundary tensor space for all three orientations. The certified common regular T germ and the positive gluing inverse therefore survive contraction with any unchanged context, including a context that reconnects two arms inside the same level-2 blob. Hence any two already-valid global endpoints differing only by local T redirections have a common full-dimensional regular region.

This proves Theorem A.

8. Canonical reconstruction

For a generic exact distribution known to arise from the class:

1. Fourier-transform the distribution.
2. Test all nontrivial splits by rank-at-most-four flattenings and reconstruct the labelled reduced bridge tree.
3. Factor each Fourier cut block into positive rank-one factors and peel the tree, obtaining projective local tensors.
4. On every local tensor, inspect bounded restrictions through seven outgoing ports. Use the exact cycle/theta invariant atlas and the new F separators to determine the generator, repair support, and port roles modulo T .
5. Use one-port and two-port probes to reconstruct every ordered segment word.
6. Reroot through an ordinary pendant path and choose the lexicographically least standard mixed-graph encoding among the valid T -orientations.

The procedure terminates because the split set and every bounded restriction deck are finite. It returns exactly the canonical observational-equivalence class. All compatible semi-directed topologies are obtained by enumerating valid ordinary T redirections and deduplicating their labelled canonical codes; no non- T topology can occur by Theorem A.

The exceptional set is the finite union of:

- proper rank/minor degeneracy varieties used by bridge and local extraction;

- proper local invariant-degeneracy varieties;
- Zariski closures of intersections with non- T topologies.

If one such intersection had full source dimension, semialgebraic stratification would produce $N \preceq_{\text{JC}} N'$, contradicting Theorem A. There are only finitely many fixed-leaf binary S_{TC} level-2 networks, so the union is proper. Dominance of the parameterization pulls it back to a proper algebraic subset of source parameter space.

9. Independent release checks

The primary seven-port implementation and a separately written reviewer agree on all 192 residual records, eight free S_4 -orbits, 1,686 canonical standard- S_{TC} completions, seven cycle rank certificates, and 65 theta rank certificates. The reviewer independently reconstructs every mixed-graph code, every complete displayed-tree monomial schema, and all 486 universal reduced witness tensors in the two genuine completion orbits.

A separate cut reviewer recomputes 547 distinct tensor-product Bernstein expansions, all 177 endpoint signs, all 421 strict wrong-split minor signs, all 32 true-bridge signatures, the seven partial-Bernstein certificates, and the four two-active-endpoint identities. A separate root reviewer checks 294,132 zero-sum complement identities and the selected/unselected retained- reticulation cases. The final release verifier reruns these independent programs, checks every dependency hash, and refuses to certify this report if any theorem dependency is missing or carries an unresolved status.

Biological consequence

Under exact infinite-data JC observations, the standard strong tree-child condition is sufficient to recover the complete level-2 history—including its bridge tree, blob generator types, and labelled descendant placements—except for the local direction of an ordinary embedded triangle. This remaining uncertainty is intrinsic: all three triangle directions share a regular JC tensor neighborhood.

The condition is sharp. Merely requiring that *some* rooting be tree-child allows the Theta pendant-transfer ambiguity, which changes the labelled semi-directed history and persists for arbitrarily many leaves.